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CREATE DATABASE RAILWAYS
USE RAILWAYS

CREATE TABLE TRAIN
(TRAIN_ID INT PRIMARY KEY,
TRAIN_NUMBER INT UNIQUE,
TRAIN_NAME VARCHAR(100),
TRAIN_TYPE VARCHAR(50),
SOURCE VARCHAR(100),
DESTINATION VARCHAR(50),
TOTAL_DISTANCE INT)


CREATE TABLE STATION
(STATION_ID INT PRIMARY KEY,
STATION_CODE VARCHAR(50) UNIQUE,
STATION_NAME VARCHAR(100),
CITY VARCHAR(100),
STATE VARCHAR(100));


CREATE TABLE TRAIN_ROUTE
(ROUTE_ID INT PRIMARY KEY,
TRAIN_ID INT,
STATION_ID INT,
STOP_NUMBER INT,
DISTANCE_FROM_SOURCE INT,
FOREIGN KEY(TRAIN_ID) REFERENCES TRAIN(TRAIN_ID),
FOREIGN KEY(STATION_ID) REFERENCES STATION(STATION_ID));


CREATE TABLE TRAIN_SCHEDULE
(SCHEDULE_ID INT PRIMARY KEY,
TRAIN_ID INT,
STATION_ID INT,
ARRIVAL_TIME TIME,
DEPARTURE_TIME TIME,
DAY_NUMBER INT,
FOREIGN KEY (TRAIN_ID),
REFERENCES TRAIN(TRAIN_ID),
FOREIGN KEY (STATION_ID),
REFERENCES STATION(STATION_ID));


CREATE TABLE COACH (
COACH_ID INT PRIMARY KEY,
TRAIN_ID INT,
COACH_NUMBER VARCHAR(10),
COACH_TYPE VARCHAR(20),
TOTAL_SEATS INT,
AVAILABLE_SEATS INT,
FARE_PER_SEAT DECIMAL(10,2,
FOREIGN KEY (TRAIN_ID)
REFERENCES TRAIN(TRAIN_ID));


CREATE TABLE PASSENGER (
PASSENGER_ID INT PRIMARY KEY,
PASSENGER_NAME VARCHAR(100) NOT NULL,
AGE INT,
GENDER VARCHAR(10),
PHONE_NUMBER VARCHAR(15),
EMAIL VARCHAR(100),
REGISTRATION_DATE DATE,
CHECK (AGE BETWEEN 1 AND 120),
CHECK (GENDER IN ('MALE', 'FEMALE', 'OTHER')));


CREATE TABLE BOOKING
(BOOKING_ID INT PRIMARY KEY,
PASSENGER_ID INT,
TRAIN_ID INT,
COACH_ID INT,
JOURNEY_DATE DATE,
BOOKING_DATE DATE,
TICKET_STATUS VARCHAR(20),
TOTAL_FARE DECIMAL(10,2),
 FOREIGN KEY (PASSENGER_ID)
 REFERENCES PASSENGER(PASSENGER_ID),
 FOREIGN KEY (TRAIN_ID)
 REFERENCES TRAIN(TRAIN_ID),
 FOREIGN KEY (COACH_ID)
 REFERENCES COACH(COACH_ID),
 CHECK (TICKET_STATUS IN
 ('CONFIRMED', 'WAITING', 'CANCELLED')));


CREATE TABLE SEAT (
SEAT_ID INT PRIMARY KEY,
COACH_ID INT ,
SEAT_NUMBER VARCHAR(20),
BERTH_TYPE VARCHAR(20),
SEAT_STATUS VARCHAR(20),
FOREIGN KEY (COACH_ID)
REFERENCES COACH(COACH_ID),
UNIQUE (COACH_ID, SEAT_NUMBER),
CHECK (BERTH_TYPE IN
('LOWER', 'MIDDLE', 'UPPER', 'SIDE LOWER', 'SIDE UPPER')),
CHECK (SEAT_STATUS IN
('AVAILABLE', 'BOOKED'))
CHECK (SEAT_STATUS IN
('AVAILABLE', 'BOOKED')));


CREATE TABLE PAYMENT
 (
PAYMENT_ID INT PRIMARY KEY,
BOOKING_ID INT ,
PAYMENT_DATE DATE,
PAYMENT_AMOUNT DECIMAL(10,2),
PAYMENT_METHOD VARCHAR(30),
PAYMENT_STATUS VARCHAR(20),
TRANSACTION_ID VARCHAR(50),
 FOREIGN KEY (BOOKING_ID)
 REFERENCES BOOKING(BOOKING_ID),
 CHECK (PAYMENT_METHOD IN
 ('UPI', 'CREDIT CARD', 'DEBIT CARD', 'NET BANKING')),
 CHECK (PAYMENT_STATUS IN
 ('SUCCESS', 'FAILED', 'PENDING')));


CREATE TABLE CANCELLATION (
CANCELLATION_ID INT PRIMARY KEY,
BOOKING_ID INT,
CANCELLATION_DATE DATE,
CANCELLATION_REASON VARCHAR(100),
CANCELLATION_CHARGES DECIMAL(10,2),
REFUND_AMOUNT DECIMAL(10,2),
REFUND_STATUS VARCHAR(20),
FOREIGN KEY (BOOKING_ID)
REFERENCES BOOKING(BOOKING_ID),
CHECK (REFUND_STATUS IN
('PROCESSED', 'PENDING', 'FAILED')),
CHECK (CANCELLATION_CHARGES >= 0),
CHECK (REFUND_AMOUNT >= 0));
```